

The data

The sampling rate of the data was 173.61 Hz. The time series you can download here are not filtered. The application of a low-pass filter of 40 Hz, as described in the manuscript, is regarded as the first step of analysis and therefore not carried out for the downloadable time series.

Files

For each set (A-E) there is a ZIP-file containing 100 TXT-files. Each TXT-file consists of 4096 samples of one EEG time series in ASCII code.

[SET A in file Z.zip](#) containing Z000.txt - Z100.txt

[SET B in file O.zip](#) containing O000.txt - O100.txt

[SET C in file N.zip](#) containing N000.txt - N100.txt

[SET D in file F.zip](#) containing F000.txt - F100.txt

[SET E in file S.zip](#) containing S000.txt - S100.txt

The Bonn University EEG dataset is a canonical, publicly available corpus for the study of electrophysiological brain signals under normal and epileptic conditions. Originally assembled and introduced by Andrzejak et al. (2001), this dataset has become a standard for benchmarking algorithms in epileptic seizure detection, neurodynamical modeling, and advanced machine learning pipelines. Each record comprises a single-channel electroencephalogram (EEG) segment of 23.6 seconds duration, sampled at 173.61 Hz and subjected to precise experimental and clinical conditions. The dataset's robust internal stratification and detailed signal statistics have underpinned advances in deep learning, geometric signal analysis, and clinical translational research.

1. Dataset Composition and Recording Protocol

The Bonn University EEG database consists of 500 non-overlapping EEG segments, each precisely 23.6 s (4097 or 4096 samples depending on export format), systematically partitioned into five sets (A-E), each containing 100 signals. The full specification is:

Set	Anatomical/Clinical Context	Electrode Type	State
A	Healthy volunteers, eyes open	Scalp (10–20)	Baseline, eyes open
B	Healthy volunteers, eyes closed	Scalp (10–20)	Baseline, eyes closed
C	Epilepsy patients, interictal, outside focus	Depth (hippocampal)	Non-seizure, outside focus
D	Epilepsy patients, interictal, inside focus	Depth (hippocampal)	Non-seizure, within focus
E	Epilepsy patients, ictal (seizure)	Depth (epileptogenic)	During seizure

All recordings are monopolar or common reference. Segments are individually selected to be approximately stationary, minimizing movement and physiological artifacts at the acquisition stage. Sets A and B capture surface EEG from five healthy individuals under two vigilance states, while C–E record depth EEG from surgical epilepsy patients during interictal and ictal episodes [1903.08100, 1903.09728, 1801.05412, 2508.12670, 2512.21170].

2. Signal Specifications and Preprocessing

All EEG segments are 4097 (sometimes 4096) samples in length ($T \approx 23.6$ s, $f_s = 173.61$ Hz). The acquisition protocol includes a band-pass filter (0.53–40 Hz) implemented at hardware or acquisition level. No additional artifact removal (such as ICA/PCA or manual rejection) is performed on the raw set; protocol details specify z-score normalization prior to network input, i.e.,

\$\$

$$x_{\text{norm}}[n] = \frac{x[n] - \mu}{\sigma}$$

\$\$

where μ and σ are the segment mean and standard deviation, respectively [1903.08100, 2512.21170, 2508.12670, 1903.09728]. In some derivatives (e.g., the UCI-repackaged Epileptic Seizure Recognition Dataset) the signals are resampled or averaged to vectors of length 178 [2111.03268].

3. Data Organization, Experiment Splits, and Class Pooling

The segmentation protocol provides several analytic possibilities:

- Three-class grouping (e.g., healthy: A+B, unhealthy: C+D, seizure: E) [1903.08100, 1801.05412]
- Two-class groupings (seizure-free vs seizure, or healthy vs seizure; sometimes O/E for eyes-closed/open) [1903.09728, 2512.21170]
- Five-class classification (A–E or Z, O, N, D, S) as per the UCI expansion [2111.03268]
- Binary groupings for specific detection challenges (e.g., AB vs E, AB vs CD, etc. [1801.05412])

Reference

[Andrzejak RG, Lehnertz K, Rieke C, Mormann F, David P, Elger CE \(2001\) Indications of nonlinear deterministic and finite dimensional structures in time series of brain electrical activity: Dependence on recording region and brain state, Phys. Rev. E, 64, 061907.](#)